

Chapter 1

Background of the Study

Introduction

From the use of mobile apps to everyday websites, users rarely think about the placement of a button or the color of a font, unless those elements fail to work. This "invisible" infrastructure is the core of User Interface (UI) design. It is the quiet engine that determines whether a digital tool feels natural or frustrating to navigate. However, as thousands of new platforms launch every day, a significant number of them remain difficult to use or simply broken. In 2023, the World Wide Web Consortium (W3C) published the Web Content Accessibility Guidelines (WCAG) 2.2, which states that web content should be accessible, perceivable, and operable. These standards are meant to ensure that individuals can interact with digital services using whatever senses are available to them. In short, inclusive design is no longer a "bonus" feature; it is a global technical requirement.

In the Philippines, this issue carries legal weight. Republic Act No. 7277, also known as the Magna Carta for Persons with Disabilities, alongside the DICT Memorandum Circular No. 004-2017, makes web accessibility a mandatory national policy. Despite these laws, the reality on the ground is different. Many local websites, including those for government services and private commerce, still fall short of these basic standards. A 2024 review of Philippine digital platforms revealed that many users with visual impairments still struggle with poor color contrast and layouts that do not

scale properly on mobile devices. The rules are in place, but the implementation is lacking.

The root of this disconnect often begins in the classroom. Students in IT and Multimedia Arts programs are taught the "theory" of design, but the real world is rarely that simple. In a typical lecture, a student might learn what a grid system is, but in a real-world job, they are faced with a client who gives vague, unhelpful feedback like "make it pop" or "it just doesn't look clean enough." Without a way to translate these human complaints into technical changes, like adjusting padding by 8 pixels or checking a luminosity ratio, junior designers often feel overwhelmed. They have the creative talent, but they lack the technical proficiency and practiced skill set that the industry requires.

The assessment of 'poor' user interface design in this study is not based on subjective aesthetic preference, but on objective technical failure. According to the 2024 WebAIM Million report, an evaluation of the top 1,000,000 websites revealed that 95.9% of home pages had detectable WCAG 2 failures. The most prevalent issue was low-contrast text (found on 81.0% of sites), followed by missing alt-text (54.5%) and empty links (44.6%). Furthermore, a 2023 industry survey by UX Design Institute noted that 68% of employers find junior designers struggle with 'pixel-perfect' execution and grid-based layouts. These statistics prove that the 'massive gap' identified in this study is a quantifiable industry-wide deficiency in technical literacy, not merely a matter of artistic taste.

This study proposes EyeCon, an interactive 2D simulation game designed to bridge this specific gap. By turning dry technical standards into a "puzzle-simulation," the proponents aim to give students a safe digital sandbox. In this simulation, they can fail, learn, and iterate without the high stakes of a real project. Rather than simply making designs visually appealing, it requires them to ensure their work functions correctly by applying the strict mathematical rules of the industry. Through this gamified approach, the study seeks to transform Accessibility Literacy from a routine checklist into a tangible, practiced skill that prepares students for the professional workforce.

Statement of the Problem

One of the most frustrating hurdles for IT and Multimedia Arts students is the massive gap between creating visually appealing work in an academic setting and developing solutions that function effectively in a professional industry environment.. While textbooks and lectures explain design rules, they do not enable students to develop the practical proficiency required for professional execution. In a real-world agency, UI design isn't just about art; it's about strict mathematical alignment, 8-point grid systems, and legal accessibility standards. When students lack a practical way to train, they often tend to prioritize aesthetics over the underlying mathematics and the WCAG 2.2 standards that professionals adhere to routinely.

Students are essentially operating without guidance in the absence of a digital sandbox that can deliver instant feedback and help them interpret unclear client instructions. Currently, there is no specialized tool that trains students to turn unhelpful

phrases like "make it cleaner" into concrete technical changes, such as adjusting contrast ratios or fixing typographic hierarchy.

To address these gaps, the proponents aim to answer the following research questions through the development of EyeCon:

1. How can *EyeCon* be developed to incorporate essential 2D tools and gameplay mechanics that simulate a professional workspace, while also integrating an instructional "Learning System"—using design trivia and real-time pop-ups—to explain the technical reasoning behind industry rules?
2. Based on the ISO 25010 software quality standard, what is the level of acceptability of *EyeCon* among UI design professionals and IT students in terms of Functional Suitability, Usability, and Performance Efficiency?
3. Is there a significant difference in the technical "Accessibility Literacy" of students before and after using the *EyeCon* simulation, as measured by a statistical comparison of pre-test and post-test scores?

Objectives of the Study

General Objectives

The proponents aim to build EyeCon as a specialized 2D simulation that helps IT and Multimedia Arts students progress beyond simply creating visually appealing work. Instead of just memorizing theory, this project provides a space where students can build the actual muscle memory needed to make technical standards and accessibility a natural part of their design process.

Specific Objectives

1. Create a professional-style 2D working environment for EyeCon that integrates tools with an educational system of trivia and pop-ups to help players grasp the technical logic for design rules while they work.
2. Gather feedback on the software quality and usability of EyeCon by having both IT students and design professionals evaluate the system against ISO 25010 standards to ensure it functions effectively as a training platform.
3. Analyze the gap between pre-test and post-test scores to see if the simulation is actually moving the needle on student learning and helping their technical accessibility literacy grow.

Significance of the Study

To Students. This research gives aspiring designers and IT students a way to implement technical design rules without the dry or confusing nature of textbook-based learning. It provides an interactive space where students can build practical skills through trial and error, moving beyond mere theory memorization to fully mastering the mathematical requirements of grid systems and accessibility standards.

To Educators. The game can serve as a digital sandbox and supplemental tool for teaching dry, technical layout rules. It allows instructors to provide instant, automated feedback to students, making it easier to facilitate hands-on learning for inclusive design and interface standards.

To Future Researchers. This study increases the exposure of gamification-based learning, which enables explaining complex techniques in a much simpler and effective way. The result of this study may serve as a foundation for proving that gamified simulation can be utilized in mastering the skills required not only for UI accessibility but also for other domains.

To De La Salle Lipa. The game helps the schools to achieve their goal of offering inclusive, high-quality education by integrating visual interface literacy, technical problem-solving, and inclusive design principles directly into their curriculum.

Scopes and Limitations

This study is about the design and development of EyeCon, an interactive 2D simulation built to turn dry design theory into an engaging, hands-on experience. This project aims to develop a digital sandbox to expand and exercise students' practical skills in line with industry standards. The focus is on precision and strategic thinking, forcing players to solve UI puzzles that rely on grid systems and inclusive design rather than artistic guesswork.

Additionally, the game EyeCon covers the technical side of UI design and digital accessibility using the Unity 6 engine with C# scripting. The design modules are based on the 8-point grid system, typographic hierarchy, and strict alignment rules. On the accessibility side, the game focuses on WCAG 2.2 standards, more specifically on teaching luminosity contrast ratios, color blind friendly palettes, and font legibility. These topics are taught in the context of a level-based junior designer simulation in which players have to both diagnose and fix “broken” interfaces using tools such as grid

overlays and contrast checkers. Players will deal with vague or confusing feedback from clients to make accurate technical adjustments, which are in turn evaluated with a mathematical grading algorithm. With the use of an interactive tool set, such as grid overlay and contrast checkers, the game makes sure that different UI concepts are explored using different logic-based tasks.

Moreover, EyeCon will be developed specifically for the PC platform in order to achieve a high level of precision and accuracy of inputs. According to Rogers, Sharp, and Preece (2023), the physical constraints and simplified ways for interaction of mobile devices can often limit the execution of complex and high-precision technical tasks. By sticking to a desktop environment, the proponents can make sure that the grading algorithm is pixel-perfect and consistent.

To keep the research focused on technical implementation, the proponents have drawn a clear line on what the game will not include. EyeCon is limited to the visual and technical stage of a project, meaning that UX research, such as the creation of personas, journey mapping, or wireframing, does not come under the scope of EyeCon. The game will also be strictly 2D. Advanced disciplines such as 3d modelling, networking, and multiplayer systems are completely excluded. Finally, the game is intended as a tool for training design logic and not as a simulation of society, which means the game does not include long-term career development and interpersonal communication beyond the interpretation of logic-based client feedback.

Definition of Terms

The proponents have defined the following technical and theoretical terms to ensure a clear understanding of the concepts discussed in this research. These definitions are provided operationally, meaning they describe how each term is specifically applied within the context of the EyeCon simulation.

8-Point Grid System.

A design principle according to which all elements of the UI are sized and spaced in multiples of eight. In the research, it is the mathematical foundation of the grading algorithm of the game in order to judge layout accuracy.

Accessibility Guidelines (WCAG) 2.2

The requirement published by W3C that will serve as the standard basis for players to follow in order to solve UI problems relating to readability, contrast ratio, and accessibility of the interface design.

Constructivism

A learning theory that promotes the idea that people learn best through the active construction of their own meaning through experience. In EyeCon, this is implemented by enabling players to learn the design rules by means of trial, error, and problem solving.

Experiential Learning

The reflection on doing that pushes the process of learning forward. This theory is applied in this research as the player is subjected to a simulated professional environment where he or she is required to practically make use of the design standards in order to fix the interface tasks labeled as broken.

Gamification

The implementation of game elements and concepts, such as a scoring system, challenges, rules, and feedback, to improve learners' knowledge retention, motivation, and engagement.

Gamified Simulation

A 2D interactive learning environment that integrates engaging gaming concepts with real-world UI design challenges. Players may address real-world design problems in a game-like environment using this simulation, which enhances their technical and UI design knowledge in an engaging and dynamic manner.

Human-Computer Interaction (HCI)

A multidisciplinary field concerned with computer technology design and human-user interaction. The study adopts the concept of HCI to make the interface of the simulation accessible and learning effective.

ISO 25010

A quality software framework that evaluates software based on IT students' and UI professionals' feedback regarding the functional suitability, usability, and performance efficiency.

Luminosity Contrast Ratio

The quantitative difference in brightness between various foreground and background elements (such as text). This is a technical aspect in the game that must be fixed by players according to the WCAG 2.2 guidelines regarding visual accessibility.

User Interface (UI) Design

The method of designing the digital interface layout that allows the player to integrate interactive elements to have a user-friendly interface.

World Wide Web Consortium (W3C)

An international organization that sets the standards and establishes rules that would apply in EyeCon for it to have a foundation to evaluate the player's work and how to execute their tasks.

Chapter 2

Review of Related Literature and Studies

This chapter presents the related literature and studies that support the development of the proposed study. These sources provide the theoretical and technical foundation of the study and help identify ideas and approaches relevant to the development of the proposed simulation game.

Technical Background

EyeCon is developed based on a particular stack of software and program languages that are selected due to their stability and applicability to the industry. The proponents selected the Unity engine to be the main structure of the project. A technical review conducted by Hussain et al. (2020) indicates that Unity is extensively known to have versatility in 2D and 3D game development with an easy to use interface that simplifies the application of an interactive environment. Their analysis shows that the Unity real-time rendering and a wide range of specialized tool support are the major advantages that can assist a developer to create responsive systems. Although they did mention that Unity may not perform tasks with large memory performance when dealing with large projects, the engine is still the most recommendable of educational simulations based on its accessibility and effective integration with C# scripts.

As the central nervous system of the sim, the game logic, the level progression and the custom scoring systems are solved with the help of C#. According to the research conducted by Bond (2022), C# is the best fit to use in Unity based projects due to its ability to provide a high-level ease to read, and the extensive control required to

access game objects. C# in the case of EyeCon does the heavy lifting of the project in terms of the implementation of the mathematical grading algorithm, which matches player layouts to 8-point grid coordinates and WCAG 2.2 contrast math. In addition, Visual Studio code is the main IDE to deal with these scripts. The fact that it is lightweight and it allows debugging it also makes it a viable option to hone the complex code that is moving the logic-based puzzles in the game.

To ensure the integrity of the code and avoid loss of data, the version control of the project is carried out by means of Git and GitHub. According to Ram et al. (2016), any given software project can not be considered as modern without the presence of a version control system because it enables the developer to track down each change, cooperate in a team, and restore the past software release in case a new script is generated that will lead to the failure of a system. Such a systematic strategy is to make sure that the development of EyeCon remains on schedule even when the proponents refine the complex UI tools and grading formulas. By storing the repository on GitHub, the proponents will be able to guarantee the developers with a secure and transparent process of development.

The visual and interactive aspects of EyeCon are developed on a set of professional design software to make the game look like a real-life working place. The pixel-perfect assets are required since the players will be expected to repair broken down publicity materials and websites. The UI elements and the graphic elements that are produced using Adobe Photoshop and Illustrator are of high-fidelity quality and would suit professional design work. According to Ambrose and Harris (2019), professional-grade tools should be used when training students on design, which is

crucial to develop their perception of visual hierarchy and technical limitations in the layout. Also, UIs are responsive through smooth animations and transitions generated with Adobe After Effects. This is a blend of art and animation features and makes sure that EyeCon is not merely a tool used to teach something, but a refined and captivating simulation.

Theoretical Framework

Experiential Learning Theory



Source: <https://experientiallearninginstitute.org/what-is-experiential-learning/>

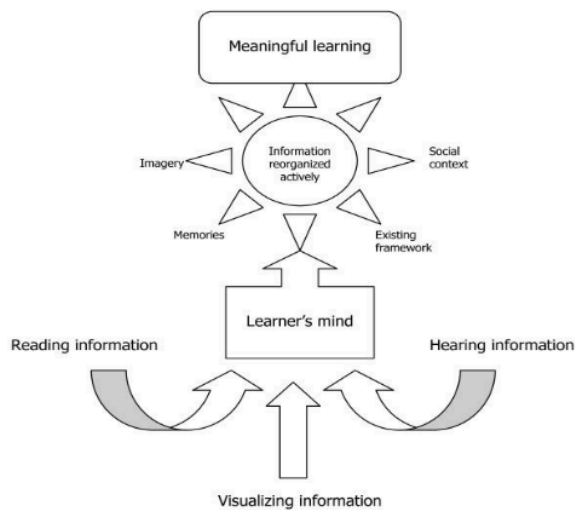
Experiential Learning Theory by David A. Kolb describes how learning occurs through experience and active participation. As Kolb said, learning is the key people use to navigate life, to grow, and to improve, with the ideal process following the four-step Experiential Learning Cycle. Naturally, the cycle is the way people learn in life, often without realizing it. Because it is composed of two pairs of opposite ways of grasping information

(Experiencing and Thinking) and transforming it into knowledge (Reflecting and Acting). Most people would have different ways in utilizing these modes, but engaging with all four leads to deeper learning.

In this study, the proposed simulation game teaches players visual interface standards and accessibility guidelines through hands-on tasks. Analyzing and fixing poorly designed UIs, receiving real-time feedback that helps players reflect and

understand design principles. This aligns with Kolb's Experiential Learning Cycle, as players experiment and refine UI design skills.

Constructivist Learning Theory



Source: <https://www.researchgate.net/publication/46287616>

Constructivist learning theory, as described by Nurhuda et al. (2023), highlights that people learn best when they actively build their own understanding through exploration, reflection, and problem-solving, rather than just memorizing facts. It focuses on hands-on, meaningful experiences that let learners discover and make sense of concepts themselves.

In EyeCon, players step into the role of a junior UI designer and are tasked with fixing broken interfaces. Instead of simply reading about design rules, they experiment with layouts, typography, hierarchy, and accessibility corrections. The game's Client Requirement Mode challenges players to interpret vague or confusing feedback and turn it into concrete design solutions mirroring real-life situations in UI/UX work. By engaging with these tasks, players actively construct their knowledge of interface standards and accessibility, turning technical concepts into practical skills they can apply outside the game.

Conceptual Foundation

Human-Computer Interaction (HCI)

Human–Computer Interaction (HCI) has moved beyond traditional usability, incorporating frameworks such as User Experience (UX), emotional design, social computing, and ethical considerations (Abelson & Smith, 2022). These frameworks emphasize designing interfaces that are emotionally engaging, socially aware, and ethically responsible. In EyeCon, a 2D simulation game for learning visual interface standards and accessibility, these principles guide the creation of interactive and meaningful learning experiences, allowing students to apply design rules in realistic contexts. By aligning with these multidimensional HCI frameworks, the game fosters both practical skills and a deeper understanding of human-technology interaction.

Review of Related Literature

Inclusive Design in Digital Interfaces

The book “Inclusive Design for a Digital World: Designing with Accessibility in Mind” by Regine Gilbert emphasizes that an inclusive design focuses on creating digital products that are accessible to as many people as possible. Keeping in mind individuals with visual, auditory, cognitive, and motor impairments. Gilbert (2019) explains that accessibility should be a priority when designing and not just an afterthought. It should be considered from the very beginning of the design process. By applying accessibility principles such as proper color contrast, readable typography, and keyboard navigation, designers can ensure that interfaces are usable for diverse groups of people. These concepts emphasize the importance of accessibility in today’s interface design and support the promotion of equal access to digital technologies.

Additionally, in the book *“Don’t Make Me Think”* (2026), it discusses the importance of usability in digital interface design. Effective interfaces should be intuitive and easy to understand so that users can accomplish tasks without confusion or unnecessary effort. Krug insists that designers prioritize clarity, visual hierarchy, and simple navigation structures to improve user experience. When interfaces are designed with usability in mind, users can interact with systems more efficiently and with fewer errors. These principles are relevant to the development of interactive systems that aim to teach proper interface design practice.

Furthermore, in the book *“The Design of Everyday Things”* by Don Norman interpreted by MacroPraxis Research Institute it explains that a human-centered design focuses on creating products that considers human behavior, perception, and cognitive processes. Norman emphasizes that good design should make functions visible and understandable so that users can easily determine how to interact with a system. The author also highlights the importance of feedback, consistency, and clear affordances in improving user interaction with digital systems. These principles serve as a foundation for many interface design standards used in modern software and application development.

Moreover, Lidwell, Holden, and Butler in *Universal Principles of Design* present a collection of design concepts that guide the creation of effective visual communication. The authors explain that principles such as alignment, hierarchy, contrast, and balance help organize information and improve readability in visual interfaces. These principles allow designers to structure digital layouts in ways that make information easier to

understand and navigate. Applying these visual design principles helps ensure that interfaces remain clear, structured, and visually effective for users.

Visual Design Standards for Interfaces

Professional systems often face visual overload, fragmented information, and limited user adaptation, increasing the risk of human error in high-pressure environments. While UI/UX design is well-studied in consumer contexts, its impact on visual ergonomics and error reduction in industrial settings remains scattered across disciplines. Research highlights ergonomic interface elements, contrast, hierarchy, spatial organization, color coding, and immediate feedback that enhance comprehension, reduce cognitive load, and improve decision-making. Emerging technologies like VR/AR and intelligent automation further demand adaptive, user-centered designs to ensure safety and efficiency (Vizcarra, Quiroz & Cornejo, 2026).

Professional UI design relies on clear visual hierarchy, spacing, alignment, typography, and color contrast to create interfaces that are both usable and accessible. Figma's UI Design Principles emphasize these standards, which help users quickly understand and navigate digital content (Figma, n.d.). Integrating these principles into EyeCon allows students to practice real-world interface rules through interactive tasks like grid overlays and contrast checks, reinforcing both design skills and accessibility awareness.

Shneiderman's Eight Golden Rules of Interface Design provide practical guidelines for creating user-friendly interfaces. They focus on keeping design

consistent, giving users clear feedback, preventing mistakes, and making it easy to undo actions. The rules also emphasize user control and reducing mental effort so users aren't overwhelmed. These principles are important for designing interfaces that are both intuitive and accessible, which is exactly what EyeCon helps students practice through interactive exercises like fixing layout errors, checking contrast, and following proper alignment and typography standards.

Simulation and Game-based Learning

Simulation immerses learners in guided replications of real-life experiences. Simulation-based learning in the health profession allows trainees and professionals to practice skills in a controlled environment using various modalities, enhancing patient safety and minimizing clinical errors. (Constantino et al. 2025)

According to Rahul Pal (2026), simulation-based learning is an interactive teaching method where learners engage with realistic scenarios to practice skills and apply knowledge safely. It bridges the gap between theory and practice by allowing decision-making, problem-solving, and experiential learning in a controlled environment. Common in fields like healthcare, engineering, and business, it improves knowledge retention, critical thinking, and preparedness, though it may require significant resources. Key elements include realistic scenarios, guided facilitation, and feedback, all promoting active and immersive learning.

ProdigyGame.com defines game-based learning (GBL) as using game elements to make learning engaging and motivating. The approach integrates structured gameplay, rewards, and challenges to enhance student focus and participation. Digital

GBL tools allow individualized practice and real-time progress tracking, improving skills like problem-solving and critical thinking. While effective in boosting engagement and learning outcomes, the blog notes that GBL should complement traditional instruction to avoid issues like excessive screen time. (Prodigy Game, 2026)

Review of Related Studies

Game-based learning

Gamified learning systems have become an efficient educational tool that incorporates components of gameplay, which also consist of tutorial content to improve the engagement of students and academic performance. According to Fang Zhang (2024), there are positive impacts of having game-based learning experiences, which are enhancing their self-efficacy and self-control when utilizing technology. The study discovered that students who are exposed to a gamified learning environment can easily adapt to the latest technology trend today. These results recommend that interactive learning games can enhance their confidence in navigating recent technologies and establishing a strong foundation of their understanding of their course materials.

Moreover, based on Al-Khayat, Gargash, & Atiq (2023), improvements in motivation, engagement, and cognitive skills can be seen in students by allowing them to use game-based learning. Students also practice their problem-solving, critical thinking, and collaboration skills as they use educational games, since students can experience an immersive environment, which helps them to develop and enhance these skills. One of the factors mentioned by the study is that the effectiveness of game

mechanics and game activities should be connected to the set learning objectives for the students.

Furthermore, research by Maria Vodenicharova (2022) emphasizes the advantages of using game-based learning tools and methods in higher education. It is reported that it reinforces knowledge acquisition, fosters strategic thinking, and enhances the decision-making skills of students. This increases the motivation, confidence, and interest of the students toward the learning material. These results strengthen the foundation of EyeCon: An Interactive 2D Simulation Game for Enhancing Visual Interface Standards and Accessibility Literacy, as the suggested system uses interactive gameplay to improve students' knowledge about the principles of interface design and accessibility standards through practical problem-solving activities.

Likewise, Alovera, Masagnay, Peña, Gabriel, and Prudente (2025) explored how improving the design of a virtual learning platform could make it easier and more enjoyable for students to use. By applying interface and experience enhancements and measuring outcomes through surveys and usability metrics like the System Usability Scale (SUS), the study found that clearer layouts, consistent navigation, and well-organized visual elements helped students perform better and feel more satisfied with the platform. This research highlights how thoughtful UI/UX design can make digital learning environments more effective and user-friendly.

Accessibility in Interface Design

The Philippine administration is making an effort towards the strengthening of e-government in the country so as to adapt to today's digital landscape. Among their

initiatives is the creation of government websites for their various offices. Previous studies have established that website credibility is perceived primarily through visual appeal and navigability. UX and UI design, however, are not as implemented on such websites despite their importance. An enhancement of the UX and UI of government websites is in order. (Arcega 2017)

Furthermore, Bokingito et al. (2025), in “Are We Inclusive? Accessibility Challenges in Philippine E-Government Websites”, found that many citizens, including those with disabilities, still face barriers when trying to access important online information and services. While some government websites have made efforts toward accessibility, gaps remain in usability, mobile access, and transactional features. This highlights the importance of designing digital interfaces that are not only functional but also user-friendly and inclusive, aligning with our research focus on how thoughtful UI/UX and visual ergonomics can reduce errors and improve the overall experience for all users.

The study UX-Design for Digital Inclusion: How to Ensure Accessibility for All Users (Kondratyuk, 2025) highlights that accessibility should be a core principle of UX design, not just a compliance measure. Inclusive design, which integrates features like keyboard navigation, high contrast modes, adaptive animations, and captions, reduces cognitive load, simplifies navigation, and improves overall usability. The research also notes benefits beyond usability, including higher user satisfaction, loyalty, and reduced support costs, demonstrating that accessibility enhances both user experience and practical outcomes.

User Interface (UI) Design Principles

Tewari (2025) proposed a framework of UX principles: Clear, Efficient, Smart, Connected, and Polished for designing digital interfaces. The study emphasizes how these principles improve usability, reduce cognitive load, and enhance user satisfaction. It also applies evaluation methods like the System Usability Scale and qualitative testing to measure effectiveness. This framework is relevant for creating engaging and user-centered digital experiences, particularly in educational and gamified platforms.

According to UI/UX Design and User Engagement (Huma, Ahmed, & Naeem, 2025) The study highlights how effective UI/UX through clear layout, intuitive navigation, and responsive interactions enhances user engagement and satisfaction on digital platforms. Mobile users are more affected by visual density, while web users prioritize navigation clarity, emphasizing the need for platform-specific design strategies.

Whaiduzzaman et al. (2023) in “Concept to Reality: An Integrated Approach to Testing Software User Interfaces” propose a hybrid evaluation framework that combines expert cognitive walkthroughs with user surveys to assess software interfaces. This approach measures usability across control, engagement, and goal domains, providing a more user-centric and quantifiable analysis than traditional methods. Application of the framework showed that integrating user feedback with expert evaluation improves UI/UX design and aligns interfaces more closely with user expectations.

Synthesis

The reviewed literature and studies focus on the emerging growth of interface design with interactive learning, wherein these methods improve digital literacy and

usability skills, which led scholars to highlight that having an inclusive human-centered design significantly impacts the development of modern interfaces. As concepts discussed by Don Norman and Steve Krug, digital interfaces must acquire effective hue contrast, cohesive typography, clear navigation, and coherent feedback; showing the top concern towards accessibility, clarity, and intuitive interaction so that digital interfaces maintain their usability for all users, ensuring inclusivity for all users. In support of this, Lidwell (2022) discusses Ben Schniederman's design framework of interface design rules and visual principles, emphasizing further the significance of having hierarchy, alignment, and structured layout in strengthening the comprehension of users and cognitive load reduction.

Moreover, research shows that gamified learning and simulation-based learning can effectively impact student engagement and knowledge retention since these learning methods enable the student to practice their critical thinking, improving their problem-solving skills as they build their confidence and enhancing their cognitive skills in such situations (Fang Zhang, 2024; Gargash et al., 2023; Vodenicharova, 2022). These realistic interactive scenarios can motivate the learners to apply decision-making and critical thinking skills and the ability to practice theoretical knowledge that they gained through game-based learning (Constantino et al., 2025; Pal, 2026). This learning method enables the students to train their decision-making and acquire practical skills without committing real-world risk, which makes the gamified learning method beneficial as a training ground for the students.

Furthermore, Arcega (2017) conducted research on e-government websites, and an accessibility evaluation by Bokingito et al. (2025) indicates that digital systems

provided to the users still have ineffective usability and design features, which led users to experience difficulties in navigating the government websites. Based on further interface evaluations by Tewari (2025), Huma, Ahmed, and Naeem (2025), and Whaiduzzaman et al. (2023), user engagement and system efficiency shall be improved as there is a proper layout, user-friendly navigation, and an inclusive interface present in digital systems. Through these findings, a gap has been identified in educational tools that teach user-interface and accessibility principles through practical interaction. Therefore, the proposed solution EyeCon: An Interactive 2D Simulation Game for Enhancing Visual Interface Standards and Accessibility Literacy intends to address the gap by incorporating standard UI design, accessibility principles, and gamified simulation learning into an immersive training environment that teaches students to acquire practical interface design skills.

Chapter 3

Research Design and Methodology

This chapter presents the research design, methodology, and procedures for data collection and analysis. It explains how EyeCon functions both as an interactive application and as the primary instrument for measuring students' technical understanding of user interface (UI) standards. The study employs a mixed-methods approach, integrating quantitative data (pre-test, post-test, and system-generated metrics) and qualitative data (interviews) to evaluate the effectiveness of the simulation.

Game Development Cycle

The development of EyeCon follows a structured five-stage cycle designed to ensure that technical design standards are accurately translated into a functional game environment. Each stage is intentionally linked to the study's research goals, serving as the process for building and validating the primary testing instrument.

Stage 1: Pre-production.

This phase focuses on research and planning, particularly the identification of technical standards such as WCAG 2.2 and the 8-point grid system, which are used as measurement criteria in the study. The proponents design game tasks and broken interface scenarios that serve as the primary assessment materials.

From a research perspective, this stage establishes the baseline competencies to be measured and defines the criteria for correct and incorrect performance.

Stage 2: Production.

During this stage, the game environment and core mechanics are developed. The system is designed to evaluate player actions based on predefined technical criteria such as layout alignment and contrast accuracy.

From a research standpoint, this stage enables the implementation of the automated data collection system, where the application records measurable variables such as placement accuracy and success rates in meeting accessibility standards.

Stage 3: Quality assurance.

The system undergoes testing to ensure that all mechanics and evaluation processes function correctly. Particular attention is given to the accuracy of the scoring and feedback system.

In the context of the study, this stage serves as instrument validation, ensuring that the data collected is reliable, consistent, and free from technical errors.

Stage 4: Launch.

EyeCon is deployed to the selected participants, specifically IT and Multimedia Arts students. Participants interact with the system by completing design tasks within the simulation.

This stage represents the primary data collection phase, where the system records performance data such as task accuracy, error frequency, and improvement across activities.

Stage 5: Post-production.

This final stage focuses on the analysis of collected data and evaluation of system effectiveness. Survey results, test scores, and gameplay data are compiled and processed.

In research terms, this stage corresponds to data analysis and interpretation, where statistical methods are applied to determine whether EyeCon leads to measurable improvement in technical UI skills.

Requirement Analysis

The proponents will undertake a requirements analysis to ensure that EyeCon is an interactive simulation and a research tool. This is the process of setting the technical and learning requirements that will be used to gauge the shift from theory to practice. This analysis in the pre-production stage will enable the proponents to ensure that the software incorporates the detailed performance indicators required for assessing the students' accessibility literacy.

Participants in this study will be IT and Multimedia Arts students at De La Salle Lipa, focusing on the age group 18-24 years old. They are selected as they are the next generation of young designers who must work with complex industry standards. As shown through general education patterns in the Philippines, a gap exists in the transition from classroom projects, where creative freedom is allowed, and the technical restraints of the industry. Therefore, the system is required to provide a controlled environment that imposes these technical constraints through logic-based gameplay,

focusing strictly on the visual implementation of the interface rather than the broader user experience research phase.

To achieve the research goals, the application is required to have some data capture and teaching functionality. EyeCon must therefore use a custom algorithm to grade and record players' progress, rather than a typical design tool. The software will be coded to record particular variables, including the precision of placement of elements based on 8-point grid positions and the frequency of success in achieving the WCAG 2.2 luminosity contrast requirements. This is critical to the research because it will give us the quantitative data required to evaluate the efficacy of the simulation in increasing technical accuracy.

Additionally, the game will need to include a client-focused workflow, to simulate real-world challenges. One requirement is to have tasks that require the player to interpret unclear feedback and apply technical changes. This is essential to assess the students' requirement analysis skills, an important element of their training. To maintain the educational nature of the system, it will provide an embedded teaching loop. This will use pop-ups and design tips to prompt the player when they're in trouble, making sure that the simulation is a constant learning experience.

Finally, to reinforce the educational foundation for the project, the proponents will seek advice from design educators and professionals. These discussions will be leveraged to fine-tune the difficulty of tasks and to ensure that the technical tasks in EyeCon are up-to-date with current industry practices. The requirements analysis, which incorporates user needs, technical standards and expert opinion, will ensure that

the final software is a valid and reliable measure of improvement in the quality of visual interface standards and accessibility literacy.

SWOT Analysis

Strength	1. The system transforms complex technical UI standards into an interactive format, improving participant engagement compared to traditional lectures. 2. It enables hands-on application of design theory, allowing the study to observe actual performance rather than relying solely on self-reported knowledge. 3. The grading system is objective and criteria-based, ensuring consistent and unbiased measurement of participants' outputs. 4. Accessibility principles (WCAG 2.2) are directly embedded into gameplay, allowing these standards to be actively assessed. 5. The system generates measurable data on participants' technical skills, such as layout accuracy and contrast compliance. 6. It provides a controlled digital environment where participants can perform tasks without external variables affecting their performance, supporting data reliability.
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Weakness	1. The game is limited to Windows PC Platform. This keeps the grading math accurate. 2. This study focuses only on UI implementation. 3. It does not cover UX research like persona building or journey mapping. 4. The scope is limited to 2D interface elements. It excludes 3D modeling and motion graphics.
Opportunities	1. EyeCon can be offered to De La Salle Lipa as a supplemental learning tool for the IT and Multimedia Arts curriculum to strengthen the students' technical UI skills. 2. The research can be expanded in the future to include levels for mobile application design or different screen resolutions. 3. The simulation may serve as a standardized assessment instrument for evaluating students' readiness for industry-level UI design tasks. 4. The collected data can contribute to future research on gamified learning and technical skill acquisition in design education.

Threats	1. The game does not guarantee engagement for all players. 2. It will likely only appeal to students interested in UI design, not the general public. 3. Web standards like WCAG could change during production. This would require updates to the grading logic. 4. Some students may find the technical rules too difficult. Those who prefer purely artistic work might struggle with the requirements.
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Design of Software and Processes

The study utilizes EyeCon as the central platform through which participants perform tasks and generate measurable outputs. The system functions as both a learning environment and a structured assessment tool. It is designed to guide participants through a series of activities while simultaneously recording their performance.

The overall process begins with participant interaction with the system, followed by automated data capture, and concludes with data extraction for analysis. The integration of these processes ensures that all relevant actions performed within the simulation contribute directly to the dataset used in the study.

Data Sources

The study is based on a combination of primary, and secondary sources. To collect primary data, three sources will be involved, including the numerical outcomes of the pre-test and post-test, the ISO 25010 software quality survey, and open-ending feedback demonstrated in the interviews with the participants. These sources collaborate to provide information to reveal whether the game is successful and the feeling of the users of the experience. The secondary sources to be considered will be official documentation of WCAG 2.2, past studies of gamified learning in the Philippines and Unity 6 tutorials. These are the documents that give technical basis to the mechanics of the game and the grading algorithm.

Survey Information and Interview Protocol

Two different instruments will be employed by the proponents to collect feedback. To gather quantitative data, first, a structured survey in accordance with the ISO 25010 standard will be conducted. This survey will have a 5-point Likert scale of the usability and performance of the game. Second, a qualitative interview guideline will be created. This protocol will not be a strict quiz, unlike the first one, but rather the open-ended questions will be posed to allow the participants to explain their own experience. Examples include asking the participants to discuss the feelings they experienced during untangling ambiguous client feedback or discussing the diagnostic tools that they found the most helpful. This combination guarantees the researchers with the knowledge of the quality of the software and the emotional involvement of players.

Data Collection

The study follows a five-stage data collection process. First, participants are recruited and oriented, and informed consent is obtained. Second, a pre-test is administered to establish baseline knowledge. Third, participants interact with EyeCon, during which performance data is automatically recorded. Fourth, participants complete the post-test and ISO 25010 survey. Finally, selected participants undergo follow-up interviews. All data is securely stored, coded, and anonymized to ensure confidentiality.

Data Analysis

The study employs a mixed-methods approach to data analysis. Quantitative data from tests and surveys are analyzed using the weighted mean and paired samples t-test to determine whether there is a significant improvement in performance after using the system.

Qualitative data from interviews are analyzed through thematic analysis, where recurring patterns and insights are identified. This helps explain how and why the system influences learning outcomes.

By combining these methods, the study provides a comprehensive evaluation of EyeCon as both a learning tool and a research instrument for improving technical UI design skills.